

HW12.5

Adam Kaderbhai

Problem 1

a.)

Explanatory variable -> Husband's age (in years) Response variable -> Wife's age (in years)

b.)

There is a positive relationship between these variables. As the Husband's age increases the Wife's age increases as well

c.)

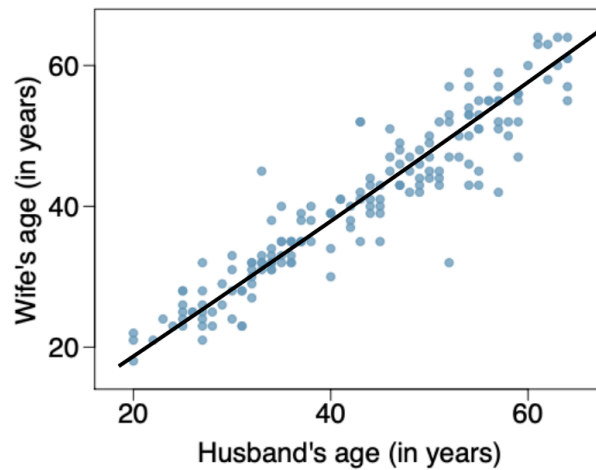


Figure 1: Line of best fit for the graph

d.)

To estimate the slope I chose two point closest to the line of best fit

(37, 38) and (41, 40)

$$(40-38) / (41-37)$$

[1] 0.5

The Estimated slope is $m = 0.5$

Problem 2

a.)

Explanatory Variable -> Family Income Response Variable -> Gift Aid from university

b.)

(71000, 21000) and (60000, 22000)

$$(22000 - 21000) - (60000 - 71000)$$

[1] 12000

$m = 12000$

Problem 3

```
garbage_weight <- read_excel("garbage_weight.xlsx")
```

a.)

```
cor(garbage_weight$METAL, garbage_weight$PLASTIC)
```

```
[1] 0.5862485
```

b.)

```
lm(garbage_weight$PLASTIC ~ garbage_weight$METAL)
```

Call:

```
lm(formula = garbage_weight$PLASTIC ~ garbage_weight$METAL)
```

Coefficients:

(Intercept)	garbage_weight\$METAL
0.6411	0.5725

$$\text{garbage_weight}\$PLASTIC = 0.641 + 0.573\text{garbage_weight}\$METAL$$

c.)

For every 1 unit of metal waste, we expect the amount of plastic waste to increase by 0.573 unit, on average.

d.)

```
metal_obs <- 2.11
plastic_actual <- 4.69

plastic_pred <- 0.641 + 0.573 * metal_obs

residual <- plastic_actual - plastic_pred

plastic_pred
```

```
[1] 1.85003
```

```
residual
```

```
[1] 2.83997
```

Problem 4

```
pokemon_sample <- read_excel("pokemon_sample.xlsx")
```

a.)

```
cor(pokemon_sample$Speed, pokemon_sample$Attack)
```

```
[1] 0.4120036
```

b.)

```
lm(pokemon_sample$Attack ~ pokemon_sample$Speed)
```

```
Call:
lm(formula = pokemon_sample$Attack ~ pokemon_sample$Speed)
```

```
Coefficients:
      (Intercept)  pokemon_sample$Speed 
        50.1866         0.4932
```

$pokemon_sample\$Attack = 50.187 + 0.493pokemon_sample\$Speed$

c.)

For every 1 unit of Speed a Pokemon has, we expect its Attack to increase by approximately 0.493 points, on average.

d.)

```
speed_obs <- 150
attack_actual <- 100

attack_pred <- 50.187 + 0.493 * speed_obs

residual <- attack_actual - attack_pred

attack_pred
```

```
[1] 124.137
```

```
residual
```

```
[1] -24.137
```